

Brazilian E-Commerce Public Dataset (Olist)

Business Intelligence Lifecycle Report, 2016-2018

Excel | Power BI | SQL Server

8 interactive dashboards, EDA and strategic recommendations

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github.com/jeistev999/Brazilian-E-Commerce-Public-Dataset

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What changed in this version. The project was rebuilt with a cleaner data model (aggregated geolocation, a dedicated date table) and redesigned Power BI dashboards. Every dashboard is now accompanied by an interpretation and a business implication, and the whole process is documented following the Business Intelligence (BI) lifecycle. Figures in this report come from the redesigned Power BI dashboards unless stated otherwise.

1. Introduction

BI LIFECYCLE | PHASE 1: BUSINESS UNDERSTANDING AND PLANNING

The purpose of this project is to analyze data from Brazilian e-commerce, based on customer orders placed through the Olist Store across different online marketplaces. The analysis combines categorical and numerical variables to understand customer behavior, product sales, seller and logistics performance, and customer satisfaction. The main goal is to produce insights that help improve sales and overall business performance by looking at all these variables together in a connected way.

The project is framed as a BI engagement: it starts from business questions, builds the data foundation to answer them, delivers dashboards for decision-makers and closes with recommendations and a monitoring plan.

Business questions

- **Sales:** which categories and states generate the most revenue, and how did sales evolve from 2016 to 2018?
- **Customers:** where are customers, how much do they spend, and do they come back?
- **Payments:** which payment methods and installment plans do customers prefer?
- **Products and sellers:** which categories sell and cancel the most, and how well do sellers dispatch orders?
- **Logistics and geography:** how long do deliveries take, what do they cost, and how do regions differ?
- **Satisfaction:** how strongly do delivery outcomes affect review scores?
- **Time:** are there seasonal peaks, and when do delays concentrate?

Scope and success criteria

The scope covers about 100,000 orders placed between 2016 and 2018. The project is considered successful if (1) the data model reproduces consistent totals across Excel, SQL and Power BI, (2) each business question is answered by at least one dashboard, and (3) each finding leads to an actionable recommendation with a measurable KPI.

2. Database

BI LIFECYCLE | PHASE 2: DATA SOURCING AND DISCOVERY

The source is the public **Brazilian E-Commerce Public Dataset by Olist** (Kaggle): <https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>. It is relational and covers the full purchase journey, from order and payment to shipping and review.

Table (CSV)	Content	Role in the model
olist_orders_dataset	Order status and key timestamps (purchase, approval, carrier, delivery, estimated delivery)	Central fact table
olist_order_items_dataset	Items per order: product, seller, price, freight	Fact (sales and freight)
olist_order_payments_dataset	Payment type, installments, value	Fact (payments)
olist_order_reviews_dataset	Review score and dates	Fact (satisfaction)
olist_customers_dataset	Customer id, unique id, city, state, zip prefix	Dimension
olist_sellers_dataset	Seller id, city, state, zip prefix	Dimension

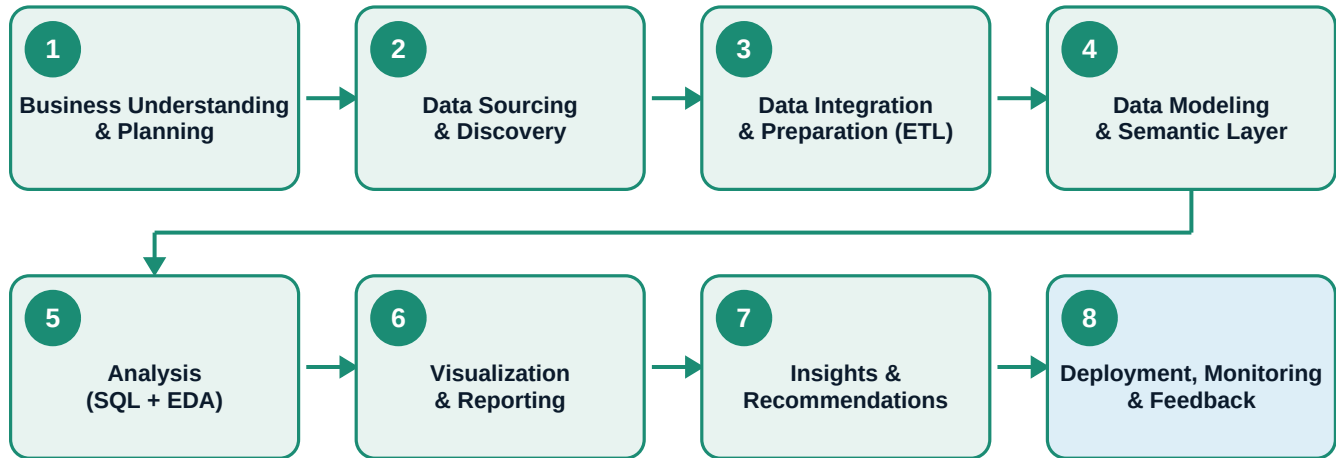
Table (CSV)	Content	Role in the model
olist_products_dataset	Category and physical attributes (weight, size)	Dimension
product_category_name_translation	Portuguese to English category names	Lookup
olist_geolocation_dataset	Zip prefix with latitude and longitude	Lookup (aggregated in v2)

Data discovery findings. During profiling, the team identified: missing values in dates, product attributes and review comments; several coordinates per zip prefix in the geolocation table; mixed data types between CSV files and the database; and the fact that the dataset covers 2016 to October 2018, with very few orders in 2016 and incomplete months at the end of the period. These findings shaped the preparation and modeling decisions that follow.

3. Methodology: the BI lifecycle applied

BI LIFECYCLE | PHASES 3 TO 6

The project followed the eight phases of the BI lifecycle shown below. Phases 1 and 2 are described in the previous sections; this section documents the technical phases and the sections that follow cover analysis, visualization, insights and deployment.



Feedback loop: monitoring results generate new business questions (back to Phase 1)

Phase	What was done in this project	Tools	Report section
1. Business understanding	Defined business questions, scope and success criteria	-	1
2. Data sourcing	Selected the Olist dataset and profiled quality issues	Kaggle, Excel	2
3. Integration and preparation (ETL)	Cleaned, typed, merged and standardized tables	Power Query, SQL Server	3.1
4. Data modeling	Built relationships, date table, calculated columns and DAX measures	Power Pivot, Power BI, SQL	3.2, 3.3
5. Analysis	SQL queries and statistical EDA	SQL Server, PivotTables	3.4, 4
6. Visualization and reporting	8 dashboards in Excel and Power BI	Excel, Power BI	5
7. Insights and action	Interpretations, recommendations and KPIs	-	6, 7
8. Deployment and feedback	Publishing in GitHub/Drive, monitoring plan	GitHub, Google Drive	8

3.1 Data integration and preparation (ETL)

Using **Power Query** (in both Excel and Power BI), the data was cleaned by fixing errors, replacing null and blank values, removing unnecessary columns and adjusting data types for dates, numbers and currency. Tables were merged using their primary keys. In SQL Server, the CSV files were imported, data types were corrected, and primary keys, foreign keys and constraints were defined to preserve integrity.

Key preparation decisions in version 2:

- **Geolocation** was aggregated by zip prefix (average latitude and longitude, primary city and state). The raw table contains many rows per zip prefix, and joining it directly multiplies rows in the fact tables.
- A dedicated **date table** (Calendario) was added to support time intelligence.
- **Derived columns** were created for delivery status (on time, late, not delivered), delivery-days bucket, delivery gap, customer type (new or returning), and installment groups.
- **Validation:** totals were reconciled against the source tables (for example, total item sales of \$13.59M and a cancellation rate of 0.63%, which matches 625 canceled orders over about 99k orders).

3.2 Data modeling

The model is built around the orders table. Items, payments and reviews connect to orders through *order_id*; customers and sellers connect through their ids; products connect to the category translation table; and a date table connects to the order timestamp. Customers also relate to the aggregated geolocation table, and a small States Brazil table supports map and state-name labels.

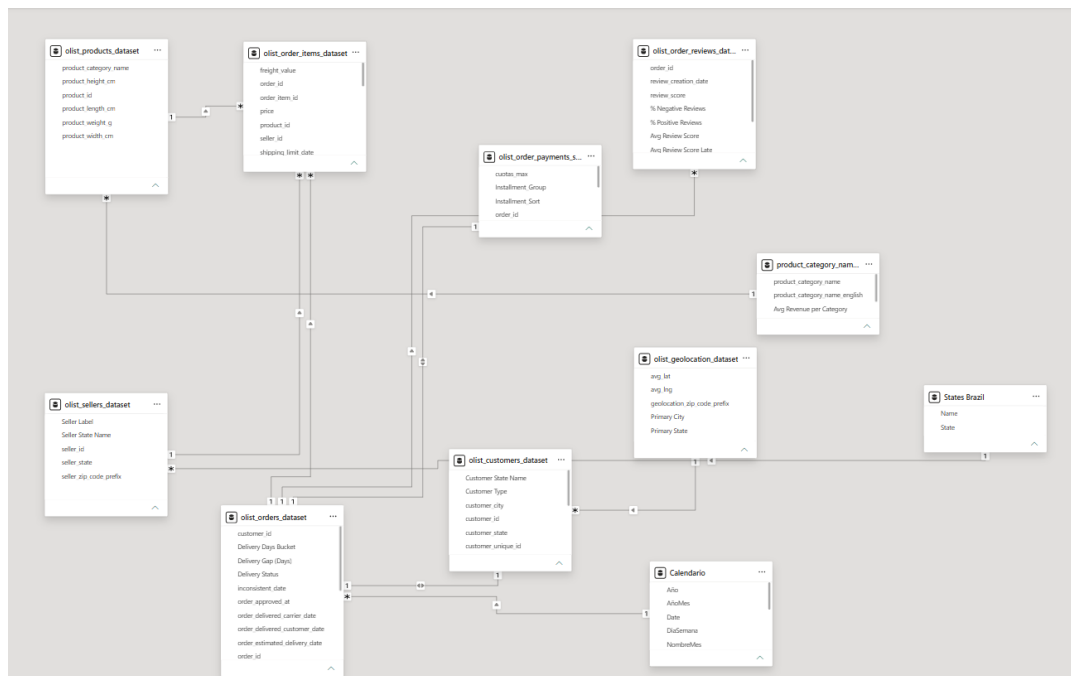


Figure 1. Relational data model in Power BI (central orders table, lookups and date table).

The model is snowflake-style: dimensions such as customers connect onward to geolocation and states. This reduces redundancy but makes filter propagation harder, which is why relationship direction and cardinality had to be reviewed several times (see Challenges).

3.3 SQL layer

In SQL Server the project includes analytical queries (sales trends by product category, revenue by customer state, order tracking by status and time), **stored procedures** that calculate total sales by month and by payment type, and **triggers** that automate validations such as flagging canceled orders and tracking late deliveries. The SQL layer acts as an independent check of the figures produced in Excel and Power BI. The full script is available in the GitHub repository.

3.4 Analysis approach

Analysis combined three techniques: SQL aggregations, PivotTables and Power Pivot measures for descriptive statistics, and Power BI visuals for pattern discovery. Section 4 describes the statistical approach and section 5 interprets each dashboard.

3.5 Why both Excel and Power BI?

Excel dashboards (PivotTables, slicers, shapes) are accessible for business users who do not have Power BI. Power BI offers cross-filtering, maps, and a better semantic model. Building both showed the strengths and limits of each tool. The redesigned Power BI version is the primary deliverable used in this report.

4. Exploratory Data Analysis

BI LIFECYCLE | PHASE 5: ANALYSIS

For each analysis area (sales, customers, payments, products, sellers, geography, reviews and time) descriptive statistics were calculated: mean, maximum, minimum, mode, median, range, standard deviation, variance, skewness, kurtosis and quartiles (Q1, Q3, IQR). The statistics are computed on **aggregated series** (totals by state, category or month), so they describe how values are distributed across those groups, not across individual orders.

How to read the statistics

- **Mean vs. median:** when the mean is much larger than the median, a few large groups pull the average up.
- **Skewness:** a *positive* value means a long tail to the *right* (a few very large values and many small ones). A negative value means a tail to the left.
- **Kurtosis:** values above 0 (excess kurtosis) indicate heavier tails than a normal distribution (leptokurtic); values below 0 indicate a flatter distribution (platykurtic).
- **IQR:** the spread of the middle 50% of values, less sensitive to outliers than the standard deviation.

Analysis	Aggregated variable	Shape of the distribution	What it tells us
Sales	Sales by month / state	Strong positive skew, heavy tails	A few months and states concentrate most of the revenue
Customers	Spend and customers by state	Strong positive skew, heavy tails	Customer base is concentrated in a few states
Payments	Payments with installments by customer state	Strong positive skew, heavy tails	Installment usage follows market size
Products	Revenue by category	Positive skew	A small number of categories dominate revenue
Sellers	Revenue by seller state	Strong positive skew, heavy tails	Seller revenue is concentrated in São Paulo
Geographic	Orders by customer state	Strong positive skew, heavy tails	Order volume is concentrated in the Southeast
Reviews	Average review score by state	Slight negative skew, near-normal tails	Satisfaction is stable across states
Temporal	Total spending by month	Slight negative skew, flat (platykurtic)	Spending grew steadily, without extreme months

Review EDA (average review score by customer state)

Statistic	Value	Statistic	Value
Mean	4.00	Q1	3.87
Median	4.05	Q3	4.11
Maximum	4.19	IQR	0.244
Minimum	3.61	Skewness	-0.85
Range	0.59	Kurtosis	-0.06
Std. deviation	0.155	Variance	0.024

Interpretation. The review score varies little across states (standard deviation 0.155, range 0.59), and the median is slightly above the mean, which produces the negative skew: a few states with lower scores pull the average down. The kurtosis of about zero means there are no unusual outliers. The maximum of 4.19 is the highest *state average*, not a single review. The order-level average shown in the Review dashboard (4.1) differs slightly from the 4.00 above because the latter is an unweighted mean of state averages.

Note on monetary EDA tables (version 1)

The monetary EDA tables of the first version (sales, customers, payments, sellers) were calculated before the data model was corrected, when joins duplicated rows and inflated totals (for example, a maximum sale of \$592M and credit card sales of \$1.25B, versus \$13.59M of total sales and \$12.54M of credit card payments in the corrected model). The distribution shapes described above remain valid and are confirmed by the dashboards (for example, São Paulo concentrates about 64% of seller revenue and the top 10 categories represent about 62% of sales). The numeric tables must be recalculated from the corrected model before being cited.

5. Dashboards and interpretation

BI LIFECYCLE | PHASE 6: VISUALIZATION AND REPORTING

Eight dashboards were built in Excel and Power BI, organized by the variables and types of data available. All Power BI dashboards share the same layout: three KPI cards, four supporting visuals and slicers for **Year**, **Primary Payment Method** and **Order Status**. Design principles applied: one business question per page, a consistent color palette, ranked bar charts for comparisons, and line charts only for time series.

Dashboard	Main question	Section
Sales	What sells, where, and how did sales evolve?	5.1
Customer	Where are customers and how much do they spend?	5.2
Payment	How do customers pay?	5.3
Product	Which categories perform, cost more to ship or cancel more?	5.4
Seller	How do sellers perform and where are they?	5.5
Geographic	How do delivery time and freight vary by state?	5.6
Review	What drives customer satisfaction?	5.7
Temporal	When do orders and delays peak?	5.8

Each subsection below presents the dashboard, the key readings, an interpretation of what the data means, and the business implication. Interpretations that are hypotheses rather than facts are marked as such.

5.1 Sales dashboard

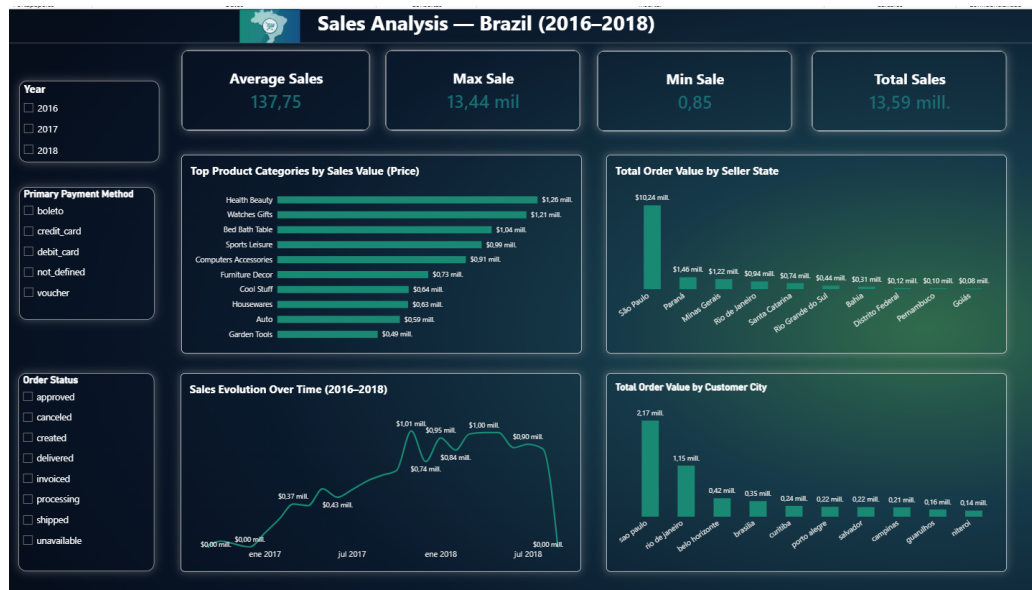


Figure 2. Sales dashboard (Power BI).

What the dashboard shows

- KPIs:** total sales \$13.59M; average sale \$137.75; maximum sale \$13.44K; minimum sale \$0.85.
- Top categories by sales value:** Health & Beauty (\$1.26M), Watches & Gifts (\$1.21M), Bed, Bath & Table (\$1.04M), Sports & Leisure (\$0.99M) and Computers & Accessories (\$0.91M). The top 10 categories add up to about \$8.5M, roughly 62% of total sales; the top 3 represent about 26%.
- Total order value by seller state:** São Paulo \$10.24M, far above Paraná (\$1.46M), Minas Gerais (\$1.22M), Rio de Janeiro (\$0.94M) and Santa Catarina (\$0.74M).
- Total order value by customer city:** São Paulo \$2.17M, Rio de Janeiro \$1.15M, Belo Horizonte \$0.42M and Brasília \$0.35M.
- Sales evolution:** sales rise from almost zero in early 2017 to roughly \$0.8M to \$1.0M per month from late 2017, with a peak near \$1.01M, and then drop to zero at the end of the series.

Interpretation

Demand is broad but not flat. Sales come from everyday consumer categories (beauty, gifts, home, sports, accessories). Revenue is spread across many categories, yet the top 10 hold more than 60%, so a portfolio strategy should protect these while testing growth in the long tail. The maximum sale of \$13.44K against an average of \$137.75 shows a long tail of rare high-ticket purchases.

Supply is far more concentrated than demand. São Paulo is the origin of most sales (\$10.24M by seller state) while customers are spread over many cities (São Paulo city is only \$2.17M by customer city). In other words, São Paulo acts as the national supplier hub, and most orders must travel between states. This links directly to the freight and delivery findings in sections 5.6 and 5.7.

Growth then plateau. The curve shows fast ramp-up during 2017 and a monthly plateau in 2018. The fall to zero at the end is a **data cutoff** (incomplete final months), not a collapse in sales. Note also that the seller-state chart reports total order value, which is higher than the item revenue reported in the Seller dashboard for the same state (São Paulo \$10.24M vs. \$8.75M); the difference is consistent with freight being included. Always check the measure definition before comparing figures.

Business implication

Protect and expand the top categories, and use bundles to lift long-tail categories. Reduce dependence on a single supplier region by recruiting sellers closer to the demand. Treat the sales plateau in 2018 as a signal that growth now requires retention and new regions, not only acquisition.

5.2 Customer dashboard

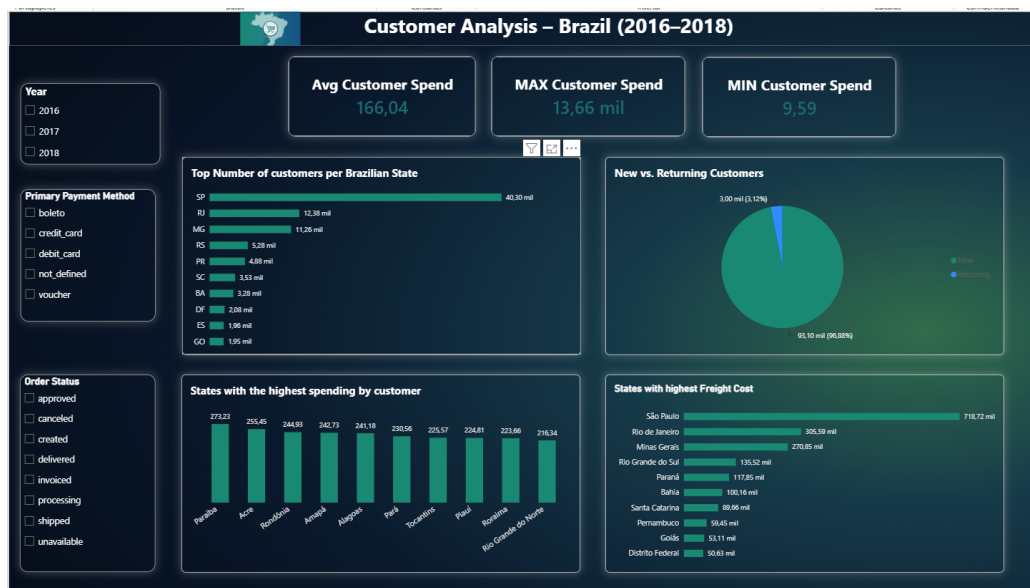


Figure 3. Customer dashboard (Power BI).

What the dashboard shows

- KPIs:** average customer spend \$166.04; maximum \$13.66K; minimum \$9.59.
- Customers by state:** SP 40.30K, RJ 12.38K, MG 11.26K, RS 5.28K, PR 4.88K, SC 3.53K, BA 3.28K, DF 2.08K, ES 1.96K, GO 1.95K. The top 3 states hold about two thirds of all customers.
- New vs. returning:** 93.10K new customers (96.88%) and 3.00K returning customers (3.12%).
- Highest spend per customer:** Paraíba \$273.23, Acre \$255.45, Rondônia \$244.93, Amapá \$242.73, Alagoas \$241.18.
- Highest freight cost:** São Paulo \$718.72K, Rio de Janeiro \$305.59K, Minas Gerais \$270.85K.

Interpretation

The customer base is a Southeast story. São Paulo, Rio de Janeiro and Minas Gerais account for roughly 66% of customers, so any national KPI is dominated by these three states.

Retention is the main weakness. Only 3.12% of customers purchased again. The marketplace behaves as a one-time-purchase channel, which makes growth depend on constantly acquiring new customers. Even a small improvement in repeat rate would add revenue without acquisition cost.

Small states spend more per customer. The ten states with the highest spend per customer are all low-volume states in the North and Northeast, and they exceed the national average of \$166.04 by roughly 30% to 65%. A plausible explanation (a hypothesis to test) is that fewer customers there buy larger or costlier baskets and pay higher freight, which inflates the order value. Total freight by state simply follows order volume (São Paulo first), so the more informative metric is freight per order, shown in section 5.6.

Business implication

Launch loyalty, referral and post-purchase programs aimed at the 96.88% of one-time buyers. In low-volume states, test targeted offers and shipping thresholds, since customers there already show a high willingness to spend.

5.3 Payment dashboard

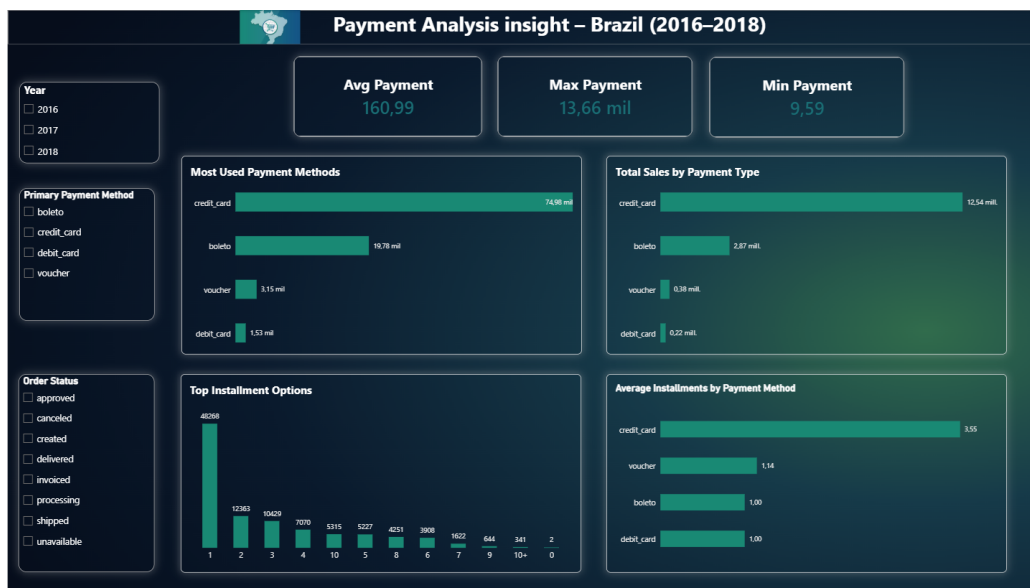


Figure 4. Payment dashboard (Power BI).

What the dashboard shows

- **KPIs:** average payment \$160.99; maximum \$13.66K; minimum \$9.59.
- **Most used methods:** credit card 74.98K, boleto 19.78K, voucher 3.15K, debit card 1.53K payments.
- **Sales by payment type:** credit card \$12.54M, boleto \$2.87M, voucher \$0.38M, debit card \$0.22M.
- **Installments:** 1 installment 48,268; 2 installments 12,363; 3 installments 10,429; 4 installments 7,070; 10 installments 5,315; 5 installments 5,227.
- **Average installments:** credit card 3.55, voucher 1.14, boleto 1.00, debit card 1.00.

Interpretation

Credit card dominates. It represents about 75% of payments and 78% of payment value. Boleto, a cash-like voucher paid at banks or lottery outlets, still carries about a fifth of payments, which shows that a meaningful share of Brazilian shoppers prefer or need non-card methods. Average value per payment is highest for credit card (about \$167) and lower for boleto (about \$145), voucher (about \$121) and debit (about \$144).

Installments are widely used on cards. The single-installment bar (48,268) is the tallest, but it includes every boleto and debit payment, which by definition have one installment. The credit card average of 3.55 installments shows that cardholders do spread payments. The first version of this report concluded that paying in installments is not widely used; the data in fact shows the opposite for credit cards.

The 10-installment spike. Ten installments is the fifth most used option, above 5 and 6 installments. This suggests that expensive purchases are pushed toward the longest plan available (hypothesis to validate by joining installments with order value). Note that total payments (about \$16.0M) are higher than total item sales (\$13.59M) because payments also cover freight.

Business implication

Keep card and installment options prominent for high-ticket categories, and promote installments in lower-volume regions to reduce the price barrier. Maintain boleto, and consider discounts or reminders to improve its conversion, because it serves customers without cards.

5.4 Product dashboard

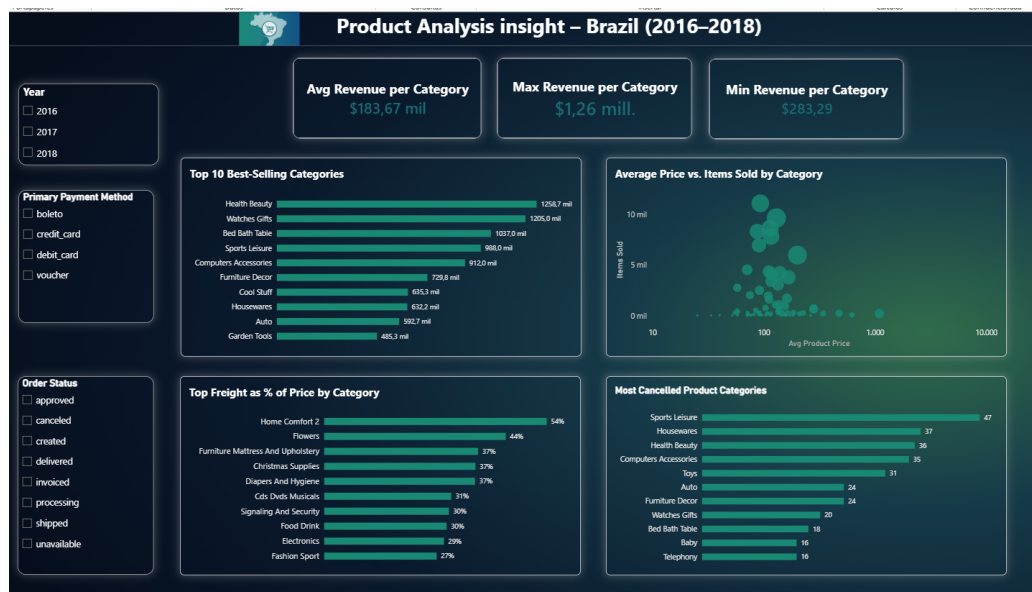


Figure 5. Product dashboard (Power BI).

What the dashboard shows

- **KPIs:** average revenue per category \$183.67K; maximum \$1.26M (Health & Beauty); minimum \$283.29.
- **Top 10 best-selling categories (revenue):** Health & Beauty, Watches & Gifts, Bed Bath & Table, Sports & Leisure, Computers & Accessories, Furniture Decor, Cool Stuff, Housewares, Auto and Garden Tools.
- **Freight as a percentage of price:** Home Comfort 2 (54%), Flowers (44%), Furniture Mattress and Upholstery (37%), Christmas Supplies (37%), Diapers and Hygiene (37%).
- **Most canceled categories (orders):** Sports & Leisure 47, Housewares 37, Health & Beauty 36, Computers & Accessories 35, Toys 31.
- **Price vs. items sold:** the bubble chart (log scale) shows that high-volume categories cluster around average prices of about \$80 to \$150.

Interpretation

Revenue is highly concentrated by category. The average category earns \$183.67K, but the best earns \$1.26M and the smallest only \$283, a gap of more than three orders of magnitude. Most volume sits in mid-priced items; expensive categories (above about \$500) sell few units.

Freight can erase the value of cheap or bulky products. When shipping equals 37% to 54% of the product price, the effective price rises sharply and conversion likely falls. Flowers and Christmas supplies are also seasonal and perishable, which raises logistic complexity.

Cancellations should be read as a rate, not a count. Sports & Leisure has the most canceled orders, but it is also a large category. Categories such as Housewares and Toys cancel a lot relative to their revenue (Toys is not in the top 10 by sales yet has 31 cancellations), which signals a higher cancellation rate. A category-level cancellation rate (cancellations divided by orders) is recommended as a KPI.

Business implication

Build bundles that pair top categories with slower ones, review the shipping policy for bulky and low-value categories (regional hubs, free-shipping thresholds, combined shipments), and investigate the root cause of cancellations in Sports & Leisure, Housewares and Toys (stock accuracy, product description, delivery promise).

5.5 Seller dashboard

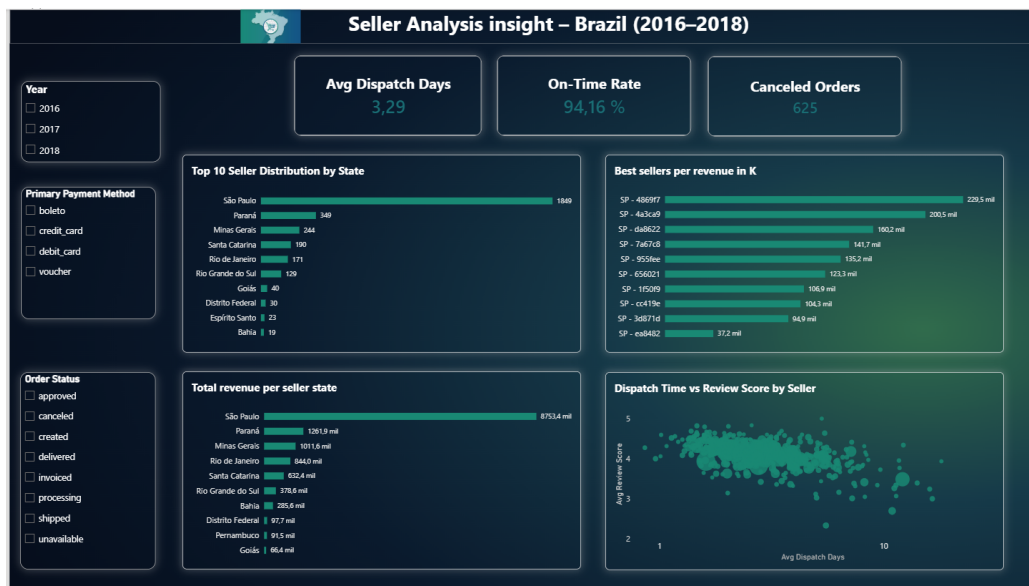


Figure 6. Seller dashboard (Power BI).

What the dashboard shows

- **KPIs:** average dispatch time 3.29 days; on-time rate 94.16%; 625 canceled orders.
- **Sellers by state (top 10):** São Paulo 1,849; Paraná 349; Minas Gerais 244; Santa Catarina 190; Rio de Janeiro 171; Rio Grande do Sul 129.
- **Total revenue per seller state:** São Paulo \$8.75M, Paraná \$1.26M, Minas Gerais \$1.01M, Rio de Janeiro \$0.84M, Santa Catarina \$0.63M.
- **Best sellers by revenue:** the ten best sellers (all in São Paulo) range from \$229.5K down to \$37.2K and together bring in about \$1.33M, close to 10% of total sales.
- **Dispatch time vs. review score:** sellers with longer average dispatch times tend to have lower average review scores.

Interpretation

Seller operations are healthy. A 3.29-day dispatch time and a 94.16% on-time rate show that most sellers hand orders to carriers promptly. Therefore, long delivery times are mostly a carrier and distance issue rather than a seller issue.

Concentration is geographic, not individual. São Paulo holds about 61% of the sellers in the top-10 states and about 64% of total item revenue (\$8.75M of \$13.59M). The ten best sellers contribute around 10% of sales, so there is no extreme dependence on a single seller, but the whole seller base sits in a single region.

Speed affects satisfaction at the seller level. The scatter plot shows most sellers clustered at 2 to 5 days of dispatch with review scores around 4 to 4.5, while sellers above about 10 days tend to score between 3 and 3.5. This is a relationship, not proof of causation, but it supports dispatch-time targets.

Business implication

Set dispatch-time service levels for sellers (with alerts above a threshold), reward top sellers, and recruit sellers in Paraná, Minas Gerais, Santa Catarina and in states far from São Paulo to shorten delivery distance.

5.6 Geographic dashboard

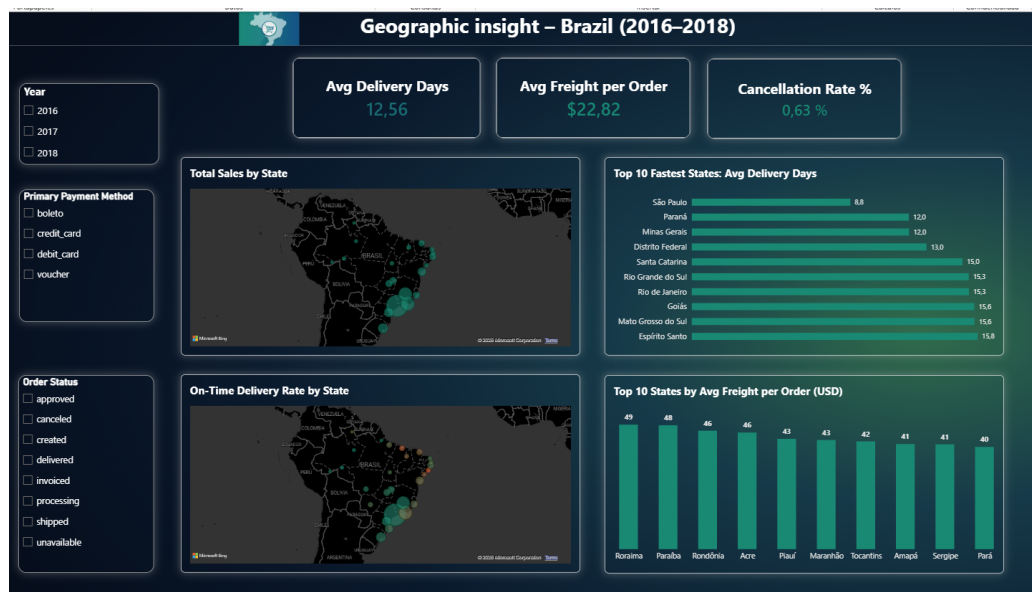


Figure 7. Geographic dashboard (Power BI).

What the dashboard shows

- **KPIs:** average delivery time 12.56 days; average freight per order \$22.82; cancellation rate 0.63%.
- **Fastest states (average delivery days):** São Paulo 8.8, Paraná 12.0, Minas Gerais 12.0, Distrito Federal 13.0, Santa Catarina 15.0, Rio Grande do Sul 15.3, Rio de Janeiro 15.3.
- **Highest average freight per order (USD):** Roraima 49, Paraíba 48, Rondônia 46, Acre 46, Piauí 43, Maranhão 43, Tocantins 42.
- **Maps:** total sales by state and on-time delivery rate by state (bubble maps).

Interpretation

Distance from São Paulo drives delivery time. São Paulo, where most sellers are, is the fastest state at 8.8 days. Even the tenth-fastest state needs 15.8 days. The overall average of 12.56 days is therefore pulled down by São Paulo's weight.

Freight roughly doubles in remote states. The ten states with the highest freight pay \$40 to \$49 per order, versus a national average of \$22.82. Combined with higher spend per customer in those same states (section 5.2), this pattern suggests that remote customers pay a significant logistics premium.

Cancellations are low. At 0.63%, cancellations are not a major problem in relative terms. The original report highlighted that canceled orders rose from 26 to 334; that increase reflects overall volume growth, so the rate should be monitored instead of the count. The maps show that sales bubbles concentrate along the Southeast and the coast, and the on-time map suggests more late deliveries in Northeast coastal states (visual reading to confirm with a state table).

Business implication

Invest in regional fulfillment or carrier partnerships for the North and Northeast, evaluate pickup points to reduce last-mile cost, and publish realistic delivery estimates by region so promises match capability.

5.7 Review dashboard

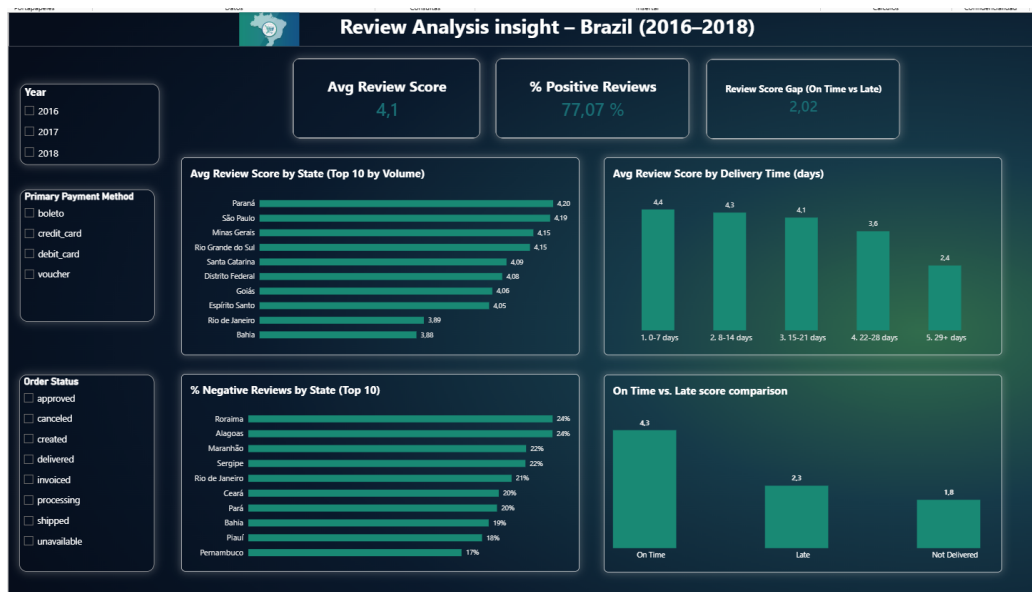


Figure 8. Review dashboard (Power BI).

What the dashboard shows

- **KPIs:** average review score 4.1; positive reviews 77.07%; score gap between on-time and late orders 2.02.
- **Average score by state (top 10 by volume):** Paraná 4.20, São Paulo 4.19, Minas Gerais 4.15, Rio Grande do Sul 4.15, ... Rio de Janeiro 3.89, Bahia 3.88.
- **Average score by delivery time:** 0 to 7 days 4.4; 8 to 14 days 4.3; 15 to 21 days 4.1; 22 to 28 days 3.6; 29 or more days 2.4.
- **On time vs. late vs. not delivered:** 4.3, 2.3 and 1.8.
- **Highest share of negative reviews:** Roraima 24%, Alagoas 24%, Maranhão 22%, Sergipe 22%, Rio de Janeiro 21%, Ceará 20%, Pará 20%.

Interpretation

Delivery performance is the single strongest driver of satisfaction. Orders delivered on time score 4.3, while late orders score 2.3 and undelivered orders 1.8. The score declines gradually up to three weeks and then drops steeply (3.6 at 22 to 28 days and 2.4 beyond 28 days), which suggests a tolerance threshold of about three weeks.

Rio de Janeiro is the largest risk. It is the second-largest market, yet it has the lowest score among the top states by volume (3.89) and 21% negative reviews. Because of its size, it likely contributes more negative reviews in absolute terms than the smaller states with higher percentages.

Negative reviews concentrate in the North and Northeast. These are the same regions with long delivery times and high freight, which gives a consistent picture: distance, cost and delay together reduce satisfaction. Overall, 77% of reviews are positive, so the base experience is good; the problem is concentrated in specific failures (delays and non-delivery).

Business implication

Treat on-time delivery as the primary customer-experience KPI. Proactively notify customers when a delay is detected, prioritize carrier performance in Rio de Janeiro and the Northeast, and apply text analysis to review comments to identify other causes of low scores.

5.8 Temporal dashboard

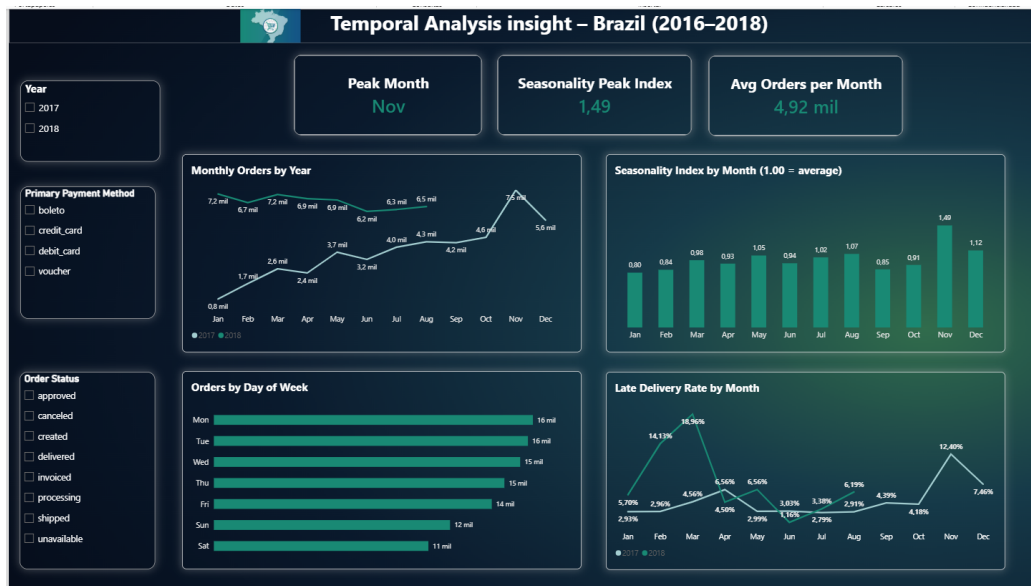


Figure 9. Temporal dashboard (Power BI).

What the dashboard shows

- KPIs:** peak month November; seasonality peak index 1.49; average of 4.92K orders per month.
- Monthly orders:** in 2017 orders climb from 0.8K (January) to 4.6K (October), peak at 7.5K in November and fall to 5.6K in December. In 2018 they stay between 6.2K and 7.2K per month up to August.
- Seasonality index by month:** November 1.49, December 1.12, August 1.07, May 1.05; lowest in January (0.80), February (0.84) and September (0.85).
- Orders by day of week:** Monday and Tuesday 16K, Wednesday and Thursday 15K, Friday 14K, Sunday 12K, Saturday 11K.
- Late delivery rate by month:** November 2017 12.40%, December 2017 7.46%, February 2018 14.13%, March 2018 18.96%.

Interpretation

Strong growth, then stabilization. Monthly orders grew roughly ninefold between January 2017 and January 2018, then stabilized. The 2016 data is too small to analyze, which is why the year slicer in this dashboard only offers 2017 and 2018. 2018 stops in August, so full-year comparisons are not possible.

November is a demand shock. The index of 1.49 means November orders were about 49% above an average month, consistent with Black Friday promotions. December keeps a high level (1.12) before the cycle returns below average in January and February.

Delays follow stress on the network. The late-delivery rate jumped to 12.4% in November 2017 and reached 14% to 19% in February and March 2018, well above the 3% to 6% seen in most months. Delays in early 2018 happened when volumes were high but not at peak, so capacity alone does not explain them; a carrier-level root-cause analysis is needed (hypothesis).

Weekly rhythm. Orders peak early in the week (Monday and Tuesday) and fall on weekends, which can guide staffing, marketing timing and carrier pick-up schedules.

Business implication

Plan inventory, seller capacity and carrier contracts ahead of November, schedule campaigns at the beginning of the week, and track the monthly late-delivery rate with an alert threshold.

6. Insights and recommendations

BI LIFECYCLE | PHASE 7: INSIGHTS AND ACTION

Each recommendation below is tied to evidence in the dashboards and to a KPI that can be monitored. Priorities are suggested according to expected impact and effort.

#	Insight (evidence)	Recommendation	KPI to track	Priority
1	Delivery time and freight rise with distance from São Paulo (8.8 days vs. 12 to 16 days elsewhere; freight \$40 to \$49 vs. \$22.82 average).	Regional fulfillment, carrier partnerships and pickup points in the North and Northeast; use AI or route-optimization tools to predict delays.	Average delivery days and freight per order by state	High
2	Late orders score 2.3 vs. 4.3 for on-time; scores fall sharply after 21 days.	Make on-time delivery the primary experience KPI; proactive delay notifications; realistic delivery estimates.	On-time rate; review gap	High
3	Only 3.12% of customers return.	Loyalty program, referral rewards, gift cards and post-purchase campaigns.	Repeat purchase rate; customer lifetime value	High
4	November index of 1.49 and late rates of 12% to 19% in stress periods.	Capacity planning for seasonal peaks (inventory, sellers, carriers); campaigns for Children's Day, Mother's Day and holidays.	Late rate by month; orders vs. capacity	High
5	Top 10 categories hold about 62% of sales; long tail underperforms.	Bundles and cross-selling that pair top categories with slow movers.	Revenue per category; bundle conversion	Medium
6	Bulky or low-value categories have freight of 37% to 54% of price; some categories cancel more than their size suggests.	Combine shipments, review shipping policy and investigate cancellation reasons by category (Sports & Leisure, Housewares, Toys).	Freight % of price; cancellation rate by category	Medium
7	Credit card averages 3.55 installments; 10-installment plan is the fifth most used.	Promote installments for high-ticket items and in low-volume states; maintain boleto for non-card customers.	Average order value; conversion by payment method	Medium
8	Sellers are concentrated in São Paulo (about 61% of top-10 states) while demand is national.	Recruit local sellers in low-supply states; set dispatch-time targets for sellers.	Sellers per state; dispatch days; distance to customer	Medium
9	Negative reviews concentrate in Rio de Janeiro and the North and Northeast.	Satisfaction surveys at purchase, shipping and delivery; text analysis of review comments.	% negative reviews by state	Medium
10	Orders peak Monday and Tuesday and fall on weekends.	Time campaigns and carrier pick-ups with the weekly rhythm.	Orders by day of week	Low

Additional recommendation: sales channel analysis. The dataset does not include the sales channel (website, WhatsApp, mobile app). Capturing it would allow optimizing user experience and investment. Physical stores, proposed in the first version, are not a priority in São Paulo given that it already has the fastest delivery; pickup points in underserved regions are a better test.

Measurement. Recommendations should be validated with controlled pilots (for example, a pickup-point pilot in two northeastern states) and compared against the KPI baselines in this report.

7. Storytelling: Project 3, Brazilian E-Commerce Analysis

BI LIFECYCLE | PHASE 7: COMMUNICATING THE INSIGHTS

E-commerce in Brazil has grown quickly, but it faces challenges in logistics, customer satisfaction and regional differences. While thinking about these issues I started to ask myself: which products sell the most and why? What payment methods do people prefer? Do customers pay with cards, with boleto, or in installments? Are customers happy, and what makes them unhappy?

To answer with objective information I used the Brazilian E-Commerce Public Dataset by Olist, which contains around 100,000 orders from 2016 to 2018 and covers the full buying process, from order and payment to shipping and review. Its relational structure was a good opportunity to apply my data skills across Excel, SQL and Power BI.

1. A market concentrated in the Southeast

São Paulo, Rio de Janeiro and Minas Gerais hold about two thirds of the 96K customers (40.3K, 12.4K and 11.3K). The seller side is even more concentrated: São Paulo alone supplies about 64% of item revenue (\$8.75M of \$13.59M). Demand is national, but supply is regional, and that mismatch explains much of what follows.

2. How customers pay

Credit card is the preferred method (74.98K payments, \$12.54M), followed by boleto (19.78K payments, \$2.87M). Cardholders use installments (3.55 on average) and the 10-installment plan is surprisingly popular, which indicates that expensive purchases are financed.

3. Distance, cost and delay

An average delivery takes 12.56 days and costs \$22.82 in freight, but the average hides large differences: São Paulo receives orders in 8.8 days while remote states pay \$40 to \$49 in freight. Overall, 94.16% of orders arrive on time and only 0.63% are canceled, so the operation works well for most customers.

4. Customer satisfaction

The average review score is 4.1 and 77% of reviews are positive. The key finding is that satisfaction depends on delivery: on-time orders score 4.3, late orders 2.3 and undelivered orders 1.8. After about three weeks of delivery, scores fall rapidly.

5. What sells

Health & Beauty (\$1.26M), Watches & Gifts (\$1.21M), Bed, Bath & Table (\$1.04M) and Sports & Leisure (\$0.99M) lead sales. Sports & Leisure also has the most cancellations (47), which warrants a root-cause analysis.

6. A retention problem hidden behind growth

Orders grew from 0.8K in January 2017 to more than 7K per month in early 2018, with a November peak driven by promotions. Yet only 3.12% of customers came back, which means growth has relied on acquiring new customers every time.

Strategic recommendations

- Improve logistics outside the Southeast and reduce freight in remote states.

- Build retention: loyalty, referral and post-purchase programs.
- Prepare capacity for November and monitor late-delivery rates monthly.
- Use bundles and targeted promotions by state and season.
- Promote installments for high-ticket items.
- Expand the seller network in regions with low supply.
- Send satisfaction surveys at purchase, shipping and delivery, and analyze review text.

8. Deployment, monitoring and feedback

BI LIFECYCLE | PHASE 8: DEPLOYMENT, MONITORING AND FEEDBACK

The final phase delivers the solution to its audience and sets up the loop that keeps it useful.

Deliverables and access

- **GitHub repository:** README, SQL script, EDA files, dashboard screenshots and this report (github.com/jeisteve999/Brazilian-E-Commerce-Public-Dataset).
- **Google Drive (read-only links):** raw dataset, cleaned dataset, EDA workbook, Excel dashboards and Power BI file (.pbix).
- **Two dashboard versions:** Power BI (primary, interactive) and Excel (alternative for users without Power BI).

Monitoring plan

If the dashboards were connected to a live source, the following KPIs would form the monitoring scorecard. The review cadence and alert thresholds below are proposals to be agreed with business stakeholders.

KPI	Current value (2016-2018)	Suggested review	Alert idea
On-time delivery rate	94.16%	Weekly	Below the historical level
Average delivery days	12.56	Weekly, by state	Increase for a state over several weeks
Late delivery rate	3% to 6% typical; up to 19% in peaks	Monthly	Above 10%
Cancellation rate	0.63%	Monthly, by category	Category above the national rate
Average review score	4.1	Monthly	Drop of more than 0.2
Repeat purchase rate	3.12%	Quarterly	Target to be defined
Freight per order	\$22.82	Monthly, by state	Rising freight in remote states
Average dispatch days	3.29	Weekly	Sellers above 10 days

Feedback loop

Stakeholder feedback and monitoring results should feed new business questions (back to Phase 1): for example, which carrier causes the delays in Northeast states, or what drives the low repeat rate. A refresh process (scheduled refresh in Power BI Service, or manual refresh in Excel) and a change log for measure definitions would keep Excel, SQL and Power BI aligned, which was one of the main difficulties of this project.

9. Conclusion

This third data analysis case study was an opportunity to work with a complex, realistic dataset and to apply the full BI lifecycle: from framing business questions, through ETL, modeling and SQL, to dashboards, recommendations and monitoring. Beyond the technical skills (data cleaning, modeling, DAX, SQL and visualization), the project reinforced a key lesson: reliable insights depend on a validated data model.

The analysis shows an e-commerce market that grew quickly in 2017, is concentrated in the Southeast on both the supply and demand side, and delivers a good experience when orders arrive on time. The main opportunities are logistics outside São Paulo, freight reduction in remote states, retention of one-time

customers and capacity planning for seasonal peaks. The recommendations are based directly on the data and come with KPIs so that their impact can be measured.

10. Challenges and lessons learned

Challenge	What happened	Resolution / lesson
Relationships and filters	The complexity of the relationships affected filters in Power BI and slicers in Excel. The model was rebuilt several times and dashboards redesigned.	Simplify the model, check cardinality and filter direction, and test each slicer against known totals.
Row duplication in joins	In the first version, joining tables at different granularities (for example, raw geolocation with several coordinates per zip prefix) multiplied rows and inflated monetary totals.	Aggregate lookups to the right grain (geolocation by zip prefix) and reconcile totals with the source before building visuals.
Loading into SQL Server	Missing values and data type inconsistencies between CSV files and table definitions.	Validate types, apply manual corrections and add constraints.
Measure definitions	Order value (includes freight) and item revenue (price only) produce different totals for the same state; some first-version KPIs were averages of state totals rather than of orders.	Document every measure (see Appendix) and label charts with the exact metric.
Interpreting statistics	Skewness sign and the unit of aggregated variables were misread in the first version (a positive skew was described as left-skewed; counts of orders were described as days).	Label variables with units, state the level of aggregation and verify results against the dashboards.
Review score range	A maximum review average of 4.19 looked inconsistent at first.	It is the highest state average, not a single review; averages of averages differ from the order-level average.
Incomplete periods	2016 has few orders and the series ends in 2018, producing artificial drops.	Flag cutoffs, restrict year slicers where data is insufficient and compare rates instead of counts.
Excel vs. Power BI differences	Small differences appeared even though both used Power Query.	Control calculations and review figures in both tools; use SQL as the third reference.
Creating eight dashboards	Large volume of data and many design decisions.	Apply a common layout and palette, and build from a clean ETL layer.

Working with Power Pivot and DAX (calculated columns and measures) was challenging but showed how important DAX is to build deeper insights.

Appendix: KPI definitions

KPI	Definition used in this report
Total sales	Sum of item prices (excludes freight).
Total order value	Item price plus freight, by order (used in the seller-state and customer-city charts of the Sales dashboard).
Average sales	Average value of a sale as shown in the Sales dashboard (\$137.75).
Total payments	Sum of payment values, which also cover freight (about \$16.0M).
Average customer spend	Average spend per customer (\$166.04).
Customer type	New or returning, derived from the customer unique id.
Average dispatch days	Days between order approval and handover to the carrier, averaged by seller.

KPI	Definition used in this report
On-time rate	Share of delivered orders delivered on or before the estimated delivery date.
Average delivery days	Days from purchase to delivery to the customer.
Freight % of price	Freight value divided by product price, by category.
Cancellation rate	Canceled orders divided by total orders (0.63%).
Positive reviews	Share of reviews with a positive score (77.07%).
Review score gap	Average score of on-time orders minus average score of late orders (2.02).
Seasonality index	Orders in a month divided by the average monthly orders (1.00 = average).

Definitions of dispatch days, delivery days and positive reviews should be confirmed against the DAX measures in the .pbix file.